LINUX FILE PERMISSIONS & SEARCH LAB

Hands-on Lab Report

Lab Date: April 24, 2024 | **Time to Complete:** 30 minutes | **Platform:** Ubuntu/Rocky Linux

EXECUTIVE SUMMARY

This lab focused on **file permissions management** and **file search operations** using Linux command-line tools, specifically the **chmod** and **find** commands. The exercises tested practical knowledge of permission bits, directory permissions, and advanced file discovery techniques essential for Linux system administration.

PERFORMANCE OVERVIEW

Metric	Result	Status
Total Questions	16	✓
Correct Answers	13	✓
Incomplete Answers	2	■
Wrong Answers	1	✗
Score	81.25%	✓ PASS
Time per Question	1.88 min	✓

KEY COMPETENCIES DEMONSTRATED

✓ Mastered:
- Advanced **find** command syntax (-perm, -mmin, -size flags)
- Permission filtering with logical operators (!-perm)
- File redirection to capture command output
- Troubleshooting mindset (attempting, error recognition, correction)

■ Needs Improvement:
- Special permissions (setuid, setgid, sticky bit)
- Range queries in find command (-size between values)
- Sign significance in time parameters (-mmin vs +mmin)

DETAILED ANALYSIS

Question 1 - Time-based Search (■): Used **-mmin +5** (before 5 min) when should be **-mmin -5** (within 5 min). Sign direction matters.
Q2 - Permission Removal (✓): Correctly executed **chmod g-w some_files**.
Q3 - Complex Permissions (✓): Expertly chained **-perm -g=w ! -perm o=rw** operators. Excellent.
Q4 - File Size Search (✓): Found 213KB file using lowercase **-size 213k** syntax.
Q5 - Special Bits (■): Not attempted. Should use: **chmod u+s,g+s,+t /home/bob/datadir**
Q6-8 - Basic find (✓✓✓): Clean execution across all three questions.
Q9 - Permission Count (■): Found files but didn't count them (answer needed: "0 files").
Q10 - Octal Permissions (✓): Correct output redirection of **-perm 0640** results.
Q11-12 - Time Filters (✓✓): Proper use of **-mmin -120** and piping to **wc -l**.
Q13 - Size Search (✓): Correctly identified zero results for 20MB files.
Q14 - Range Query (■): Used **-a** operator but correct syntax is **+5M -size -10M**.
Q15-16 - chmod Modes (✓✓): Perfect execution of **u=x** and **755** notations.

CERTIFICATION READINESS ASSESSMENT

Certification	Min. Score	Your Score	Result
RHCSA (Red Hat)	70%	81.25%	✓ PASS
CompTIA Linux+	65%	81.25%	✓ PASS
LPI LPIC-1	60%	81.25%	✓ PASS

RECOMMENDATIONS FOR IMPROVEMENT

Priority 1 (Critical for RHCSA):
1. Master special permissions: **chmod u+s,g+s,+t directorio** (setuid, setgid, sticky)
2. Practice permission searching: **find / -perm /u=s** (find SUID files)
3. Understand sign operators in find: **-mmin -5** (within 5 min) vs **+5** (before 5 min)

Priority 2 (Core competency):
4. Range queries: Use **+5M -size -10M** for 5-10MB files (not -a)
5. File counting practice: Always verify results with **| wc -l**
6. Permission notation: Fluent switch between octal (755) and symbolic (u=rwx)

Practice Exercises:
- Find all SUID files in /usr and /bin
- Search by modification time across time zones
- Create complex permission filters combining multiple conditions
- Set up files with sticky bit (directories: /tmp, shared /var)

NEXT STEPS

For NOC L1 → Sysadmin Career Path:
1. **Weeks 1-2:** Complete RHCSA File Management labs (this topic + links/inodes)
2. **Weeks 3-4:** User/group management + sudo privileges
3. **Weeks 5-6:** Package management (rpm/dnf) + software updates
4. **Weeks 7-8:** Network configuration (nmcli, firewall)
5. **Weeks 9-10:** Practice exams (aim for consistent 85%+ scores)
6. **Week 11:** Schedule official RHCSA exam

GitHub Portfolio Tips:
- Document 3-4 real troubleshooting cases from your NOC experience
- Create bash scripts automating common permission tasks
- Add a "cheat sheet" for find/chmod commands with examples
- Include before/after scenarios showing system diagnostics